# Ex.No. 2: Implementing TDMA Slot Allocation in GSM.

**Objective:**

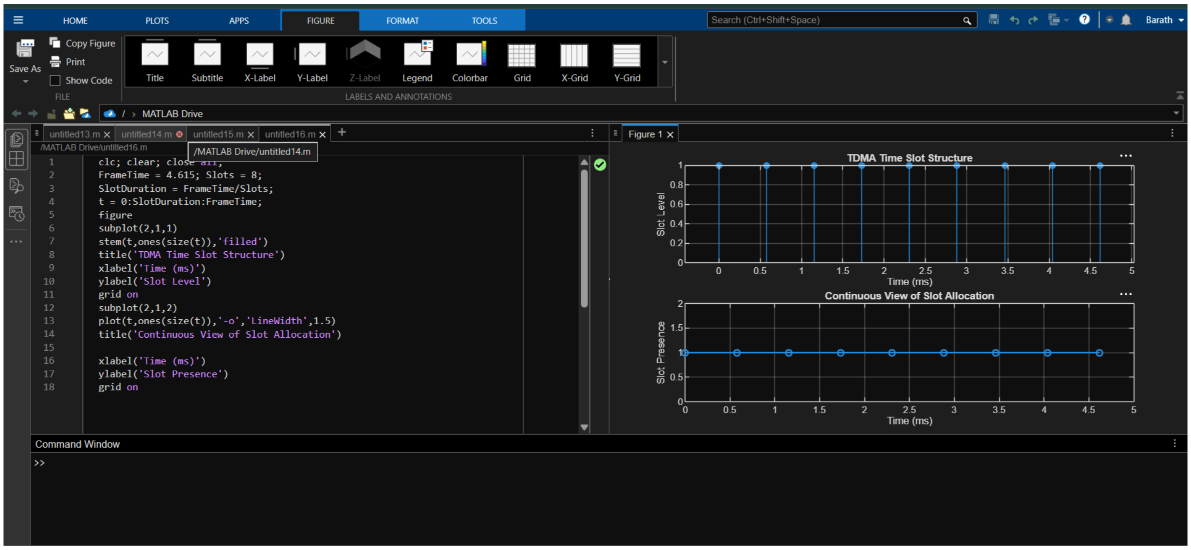
1. To calculate the Time Slot Duration in a GSM TDMA frame by dividing the total frame time into equal time slots. This helps in understanding the time structure of GSM communication.

2. To determine the Frame Utilization (%) based on the number of allocated time slots compared to total available slots. This evaluates the efficiency of TDMA resource usage.

3. To compute the Number of Allocated Time Slots for multiple users in the GSM system. This ensures proper scheduling and interference-free communication.

**Objective 1: Matlab Code and Output**

```
clc; clear; close all;

FrameTime = 4.615; Slots = 8;

SlotDuration = FrameTime/Slots;

t = 0:SlotDuration:FrameTime;

figure

subplot(2,1,1)

stem(t,ones(size(t)),'filled')

title('TDMA Time Slot Structure')

xlabel('Time (ms)')

ylabel('Slot Level')

grid on

subplot(2,1,2)

plot(t,ones(size(t)),'-o','LineWidth',1.5)

title('Continuous View of Slot Allocation')
```




**Objective 2: Matlab Code and Output**

```
clc; clear; close all;

TotalSlots = 8;

AllocatedSlots = 6;

FreeSlots = TotalSlots - AllocatedSlots;

Utilization = (AllocatedSlots/TotalSlots)*100;

disp('Frame Utilization (%)')

disp(Utilization)

figure

% Graph 1: Simple Line Plot (SAFE - no bar function)

subplot(2,1,1)

plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2)

title('GSM Frame Utilization (Allocated vs Free)')

xlabel('Slot Type Index (1=Allocated, 2=Free)')

ylabel('Number of Slots')

grid on
```

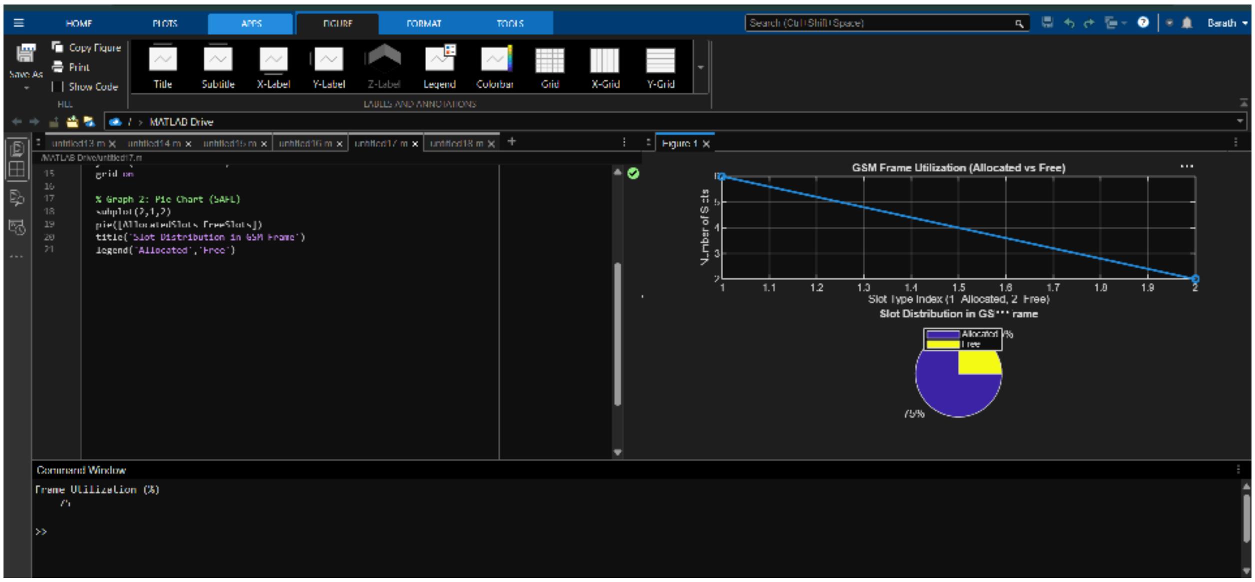
% Graph 2: Pie Chart (SAFE)

subplot(2,1,2)

pie([AllocatedSlots FreeSlots])

title('Slot Distribution in GSM Frame')

legend('Allocated','Free')




**Objective 3: Matlab Code and Output**

clc; clear; close all;

Users = 1:8; % Total users

TotalSlots = 5; % Available slots

Alloc = zeros(1,8); % Initialize allocation

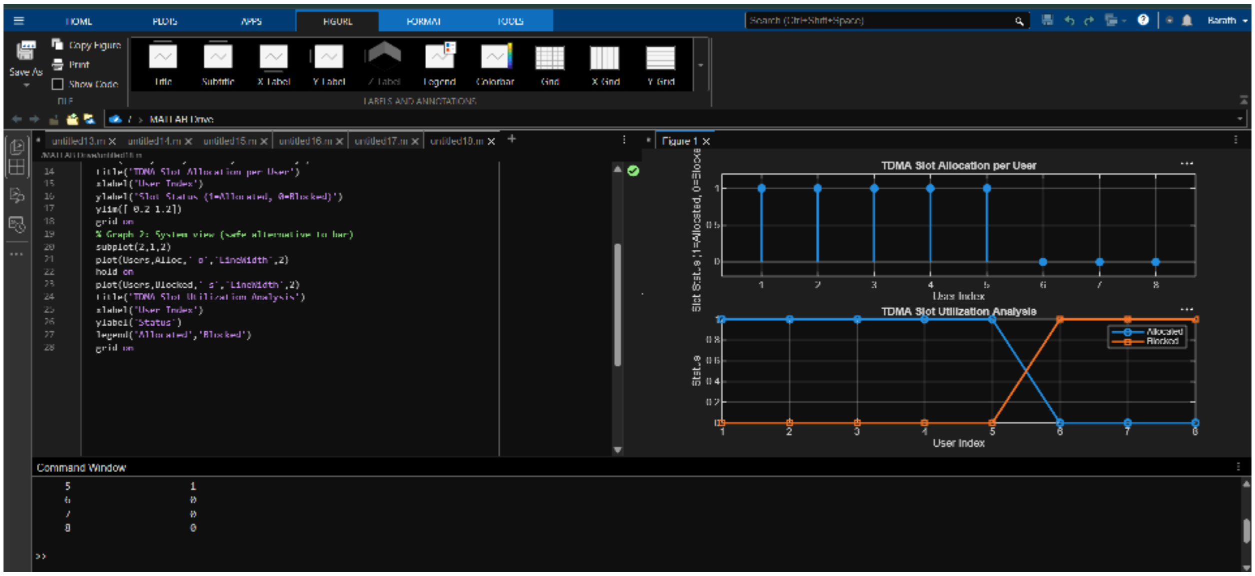
Alloc(1:TotalSlots) = 1; % Assign slots to first users

Blocked = 1 - Alloc; % Remaining users blocked

disp('User Wise Slot Allocation')

disp(table(Users',Alloc','VariableNames',{'User','Status(1=Allocated,0=Blocked)'}))

```
figure
% Graph 1: Allocation per user
subplot(2,1,1)

stem(Users,Alloc,'filled','LineWidth',2)
title('TDMA Slot Allocation per User')
xlabel('User Index')
ylabel('Slot Status (1=Allocated, 0=Blocked)')
ylim([-0.2 1.2])
grid on
% Graph 2: System view (safe alternative to bar)
subplot(2,1,2)
plot(Users,Alloc,'-o','LineWidth',2)
hold on
plot(Users,Blocked,'-s','LineWidth',2)
title('TDMA Slot Utilization Analysis')
xlabel('User Index')
ylabel('Status')
legend('Allocated','Blocked')
grid on
```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LABELS AND ANNOTATIONS
MATLAB Drive
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untitled15.m
untitled16.m
untitled17.m
Figure 1
title('TDMA Slot Allocation per User')
xlabel('User Index')
ylabel('Slot Status (1=Allocated, 0=Blocked)')
ylim([ 0.2 1.2])
grid on
% Graph 2: System view (safe alternative to bar)
subplot(2,1,2)
plot(Users,Alloc,' o','LineWidth',2)
hold on
plot(Users,Blocked,' s','LineWidth',2)
title('TDMA Slot Utilization Analysis')
xlabel('User Index')
ylabel('Status')
legend('Allocated','Blocked')
grid on
TDMA Slot Allocation per User
User Index
TDMA Slot Utilization Analysis
Allocated
Blocked
Status
User Index
Command Window
5 1
6 0
7 0
8 0
>>